



Advanced Cisco Networking Academy: Setting up Local AI with Web UI Integration

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Purpose

This lab contributes to building a basic understanding of how to implement AI into the workspace. This lab additionally helps to develop skills in creating and implementing containers, using API keys to communicate between containers. Basic skills such as implementation of Local AI, Containers, and APIs help to build a foundation of basic skillsets that will help in future endeavors such as training a Local AI to handle specific tasks.

Background

Since the advent of AI in 1950 its development has grown at an exponential pace, from only being able to talk to its users to becoming an agent that can tackle mundane tasks, increase work efficiency, and cut operating costs for small and large businesses alike. AI has become so advanced to the point where consulting firms like Gartner along with 79% of corporate strategists agree that AI integration in businesses will be vital and necessary for businesses to survive against competitors who are already using AI in the coming years. The reason AI now serves such a critical role in businesses is due to a few key benefits. The first being increased work efficiency and productivity, through the automation of repetitive tasks, streamlining workflows, providing data driven insights which according to management consulting firm McKinsey reduces the menial workload that takes up to 60 to 70% of an employee's time. Second, reducing costs through streamlining and automation of repetitive tasks enabling a company to have AI handle tasks previously an employee would need to handle, additionally businesses with AI integration could predict future Maintenance expenses and manage power more efficiently reducing costs. Lastly, AI can help a business or employee make decisions more effectively by analyzing large data sets at once, to provide insight for an employee to revise and make a final decision.

Additionally, a survey by Deloitte showed that 59 percent of executives reported that AI enabled them to get more actionable insights from analytics they were tracking. Due to these benefits that AI can provide to businesses, there is a need for businesses to utilize AI to survive among their competitors. The first step with integrating AI would be to have a local AI on-site to consult with, but this would require hardware resources onsite to allocate specifically just for the AI, posing the question why business owners cannot simply use a web-based model like Chat-GPT from OpenAI.

Web-based models like Chat-GPT make a serious flaw when integrating AI into the workforce. For instance, if a company employee needs to use AI to analyze some sensitive company data, the moment that data is inputted into the web model becomes a part of its code held in OpenAI datacenters. Although most models have restrictions on sharing user information with other users, people have found a way to jailbreak models enabling them to get the AI to tell them any information they want. Also, if data

from OpenAI ever gets leaked, company information would also be available for malicious actors. An alternative to having dedicated hardware on-site is to have it off-site using other companies' hardware. This would be a cloud-based AI that companies can use without having to have hardware resources on-site. Cloud-based AI also has its own security and performance drawbacks. The first and most important being the security risks, having a cloud-based AI that all employees in a company can use is susceptible to man-in-the-middle attacks as people can steal potentially sensitive information that you are sending to cloud servers through these attacks raising the risk of leaking company/user information. Additionally, cloud servers are a frequent target for hackers and have been breached multiple times in famous incidents like the 2019 Capitol One breach and the more recent 2024 change Healthcare incident. Local AI eliminates these security risks of passing through third party servers and having data stolen by having everything on-site. Cloud-Based AI also has more latency compared to Local AI as to get a response from a Cloud-based AI you need to have constant connectivity to the cloud server. Whereas using Local AI you will always have connectivity regardless of internet as the hardware is on-site, you can get instant responses. Local AI is also more cost efficient in the long term as having cloud infrastructure costs money if it is being used. For example, companies can avoid recurring fees associated with cloud storage, data transfer, and API requests by investing in a local AI. But there are some drawbacks to Local AI that should be known and considered when implementing AI into businesses. The main drawback of local AI is that you need dedicated hardware and sufficient quality of hardware to run the AI depending on how complex the model a business needs to run. The second drawback is related to the first one, as to upgrade a local AI you would need to invest in more hardware. But even considering you need dedicated hardware to effectively integrate local AI into your business, the security and long-term cost benefits provided by local AI far outweigh the drawbacks of the required hardware.

When setting up local AI for businesses, containers play a vital role in protecting the business network. Without containers, businesses increase the risk of data breaches, intellectual property theft, and operational chaos. Without the use of isolated environments like containers malicious AI models, and malicious actors utilizing prompt injection can manipulate the AI into exploiting vulnerabilities in the host's operating system, potentially leading to the AI taking over the host and other connected devices on the network. Containers work to prevent these risks of running Local AI as containers are an isolated environment from the host's operating system. Containers do this by creating an abstraction at the application layer that packages code and dependencies that are running in an isolated process in a user space. These features allow containers to work with any host or operating system, so once it is

configured, it does not need to be configured again. This way these AI containers not only prevent security risks but also make portable and consistent between system. An attack that malicious actors use, exclusive to LLMs/AIs, is prompt injection. Prompt injection is a method of manipulating an AI through malicious prompts to extract data from the host, remotely execute code, or transmit malware. Containers help prevent this by restricting the privileges of access to the hosts and network files, restrictions of what it can and cannot do (ex: non-root privileges), and isolation of system calls preventing what the AI can ask the operating system to do. The benefits of the container make it a necessity when running local AI so that it does not interfere with operations and impose security risks on a business. Tools like Docker have a friendly user interface that enables people without knowledge of command line commands to create and utilize containers that can run local AI.

Choosing the correct AI for the workplace is particularly important as choosing an incorrect model may result in having a corrupt model, a model that is too slow, or a model that does not give good insight and quality responses. Choosing the correct model depends on the hardware you have available. Models have something called parameters which help an AI learn and generate better responses. In general, more parameters mean better responses, but too much will require more computational resources that slow down response times. This is why finding a good model depending on the hardware a company has is important. Models can range from two billion parameters to a few trillion parameters (GPT-5), but for most use cases seven billion parameters provide quick and accurate responses, and 70 billion parameters for more complex reasoning tasks. To see what model a piece of hardware can run depends on the GPU memory with the general formula being: $M = (P * 4B * (32/Q) * 1.2) / 1e9$, where M is GPU memory in Gigabits, P is the number of parameters, Q is the bit precision (e.g., 16 or 32), and 1.2 accounts for 20% overhead. For example, a 70 billion parameter model recommended amount of GPU memory would be 168 gigabits.

User interfaces are a vital part of integrating a local AI within the workforce as most employees and workers do not know any command line prompts to effectively use the AI. With user interfaces like Open WebUI company employees can effectively start to use AI without needing to take the time to learn commands to increase their productivity and start using AI. Interfaces like Open WebUI also allow for multiple different AIs to be accessible through one user interface so that employees can easily switch between different AIs that are optimized for different tasks in the workplace. A customized web user interface can also allow employees to control how the AI gives responses, allowing for the manipulation of variables like Temperature, Top-P, Top-K which control how creative or down to the point the responses are. The benefits of a

User Interface allow for easy onboarding and quick integration of local AI within a company so that businesses do not fall behind in the new era of AI.

Lab Summary

First, to simply have a local AI without a WebUI I downloaded Ollama and then chose an appropriate model and downloaded it.

To create an AI with a WebUI, I used a tool called Docker to create containers to safely run AI models within an isolated environment. From Docker, I downloaded an AI to run inside of the container. After creating the container, I connected it with Open WebUI through an API to ease the use of AI by allowing access through a web user interface.

I also created a separate Web UI using LM Studios. First downloading Gemma on LM studios running it and making it accessible in the developer's tab. Then Prompting Claude to write a python script after giving it the API from LM studios and running it.

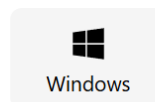
Lab Commands

Setting Up Local AI with Ollama

Download Ollama

Go to ollama.com/download/windows and download the correct version for your operating system.

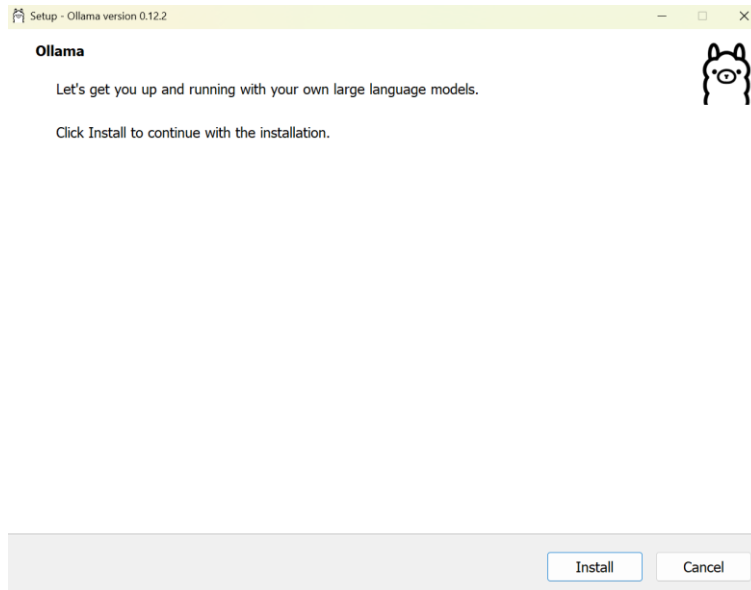
Download Ollama



Download for Windows

Requires Windows 10 or later

Follow the directions to finish installing



Open Ollama and choose an appropriate model and download

Select a model ▾ +



Once the model is downloaded verify that it works

hi

How's it going? Is there something I can help you with or would you like to chat?



Local AI with Web UI using Docker

Download Docker at <https://www.docker.com/products/docker-desktop/>

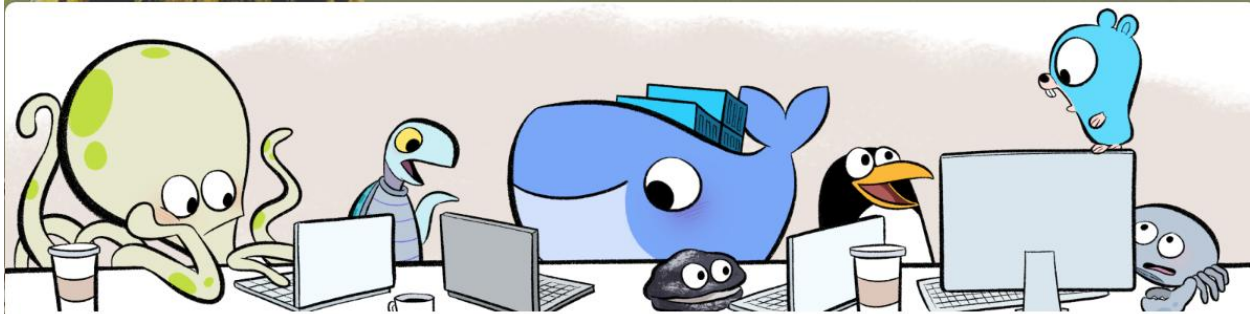
The #1 containerization software for developers and teams

Streamline development with Docker Desktop's powerful container tools.

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Allow Docker to make changes and agree to the User Agreement



Docker Subscription Service Agreement

By selecting **accept**, you agree to the [Subscription Service Agreement](#), the [Docker Data Processing Agreement](#), the [Data Privacy Policy](#) and the [Docker AI Supplemental Terms](#).

Commercial use of Docker Desktop at a company of more than 250 employees OR more than \$10 million in annual revenue requires a paid subscription (Pro, Team, or Business). [See subscription details](#)

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Accept

Close

Navigate to the blue search bar located at the top of Docker and search “Ollama/Ollama” and pull the latest image.

ollama

Ask Gordon

Images (51)

Containers (1)

Volumes (0)

Extensions (0)

Docs (6)

Hub images (50)

Hub private repos (0)

Local images (1)

 [ollama/ollama](#)

10M+ · 1.4K

Tag latest ✓

Pull

Run

Once the Image is pulled go to Images and click the run button and copy the optional setting below:

 **Run a new container**
ollama/ollama:latest

Optional settings

Container name
Ollama

A random name is generated if you do not provide one.

Ports

Enter "0" to assign randomly generated host ports.

Host port
11434 :11434/tcp

Volumes

Host path ... Container path +

Environment variables

Variable Value +

Cancel

Run

Next pull a Open WebUI image using this command on your terminal:

docker pull ghcr.io/open-webui/open-webui:git-d1c9555-cuda

```
Microsoft Windows [Version 10.0.26100.6584]
(c) Microsoft Corporation. All rights reserved.

C:\Users\kacin>docker pull ghcr.io/open-webui/open-webui:git-b38d59d-cuda
git-b38d59d-cuda: Pulling from open-webui/open-webui
4f4fb700ef54: Download complete
480eecd088ec: Download complete
244621eeded6: Download complete
630310d39a43: Pulling fs layer
53a5bc32867c: Downloading [=====>] 31.46MB/39.62MB
17dfcfa0db85: Download complete
ddf5ee7c7787: Pulling fs layer
a0c696de751f: Download complete
6a53e6668267: Download complete
e8808a2bb780: Downloading [==>] 4.194MB/91.03MB
659036234d55: Downloading [=====>] 1.049MB/3.516MB
84a423c872e8: Download complete
d107e437f729: Pulling fs layer
ce4392d98604: Pulling fs layer
1ce95d9ae66c: Pulling fs layer
|
```

(Notice the command is slightly different due image updates. For the latest image go to:

<https://github.com/open-webui/open-webui/pkgs/container/open-webui>)

Navigate to Images and run the Open WebUI Image and copy these configurations:



Run a new container

ghcr.io/open-webui/open-webui:git-f472143-cuda126

Optional settings



Container name

Open WEBUI

A random name is generated if you do not provide one.

Ports

Enter "0" to assign randomly generated host ports.

Host port

8080

:8080/tcp

Volumes

Host path



Container path



Environment variables

Variable

API

Value

http://host.docker.internal:11434



Cancel

Run

Notice that to connect the Web UI with Ollama I added an environmental variable with the address of the Ollama container. Setting the port number allows you to know the port mapping and access your AI more easily.

Verify the AI by typing Localhost:8080 into a browser and start prompting

What are 5 creative things I could do with my kids' art? I don't want to throw them away, but it's also so much clutter.

OI **llama3.1:8b** Today at 1:50 PM

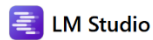
The perpetual conundrum of the proud parent! Don't worry, I'm here to help you find some creative ways to preserve and display your little ones' masterpieces while keeping the clutter at bay.

Here are 5 ideas to get you started:

1. **Art Portfolio:** Create a portfolio for each child (or even a shared one) where you can store their artwork over time. Use a binder, folder, or a beautifully designed notebook with pages dedicated to different themes or periods of art-making.

LM Studios AI + WebUI

Download LM Studios at <https://lmstudio.ai/>



Local AI, on your computer.

Run local AI models like `gpt-oss`, `Qwen`, `Gemma`, `DeepSeek` and many more on your computer, privately and for free.

 Download for Windows 0.3.27

LM Studio is free for home and work use • [terms](#)

Select any option when prompted to choose your level

[← Back](#)

[Skip →](#)

Choose your level



User

I'm just getting started with AI



Power User

I know what I'm doing

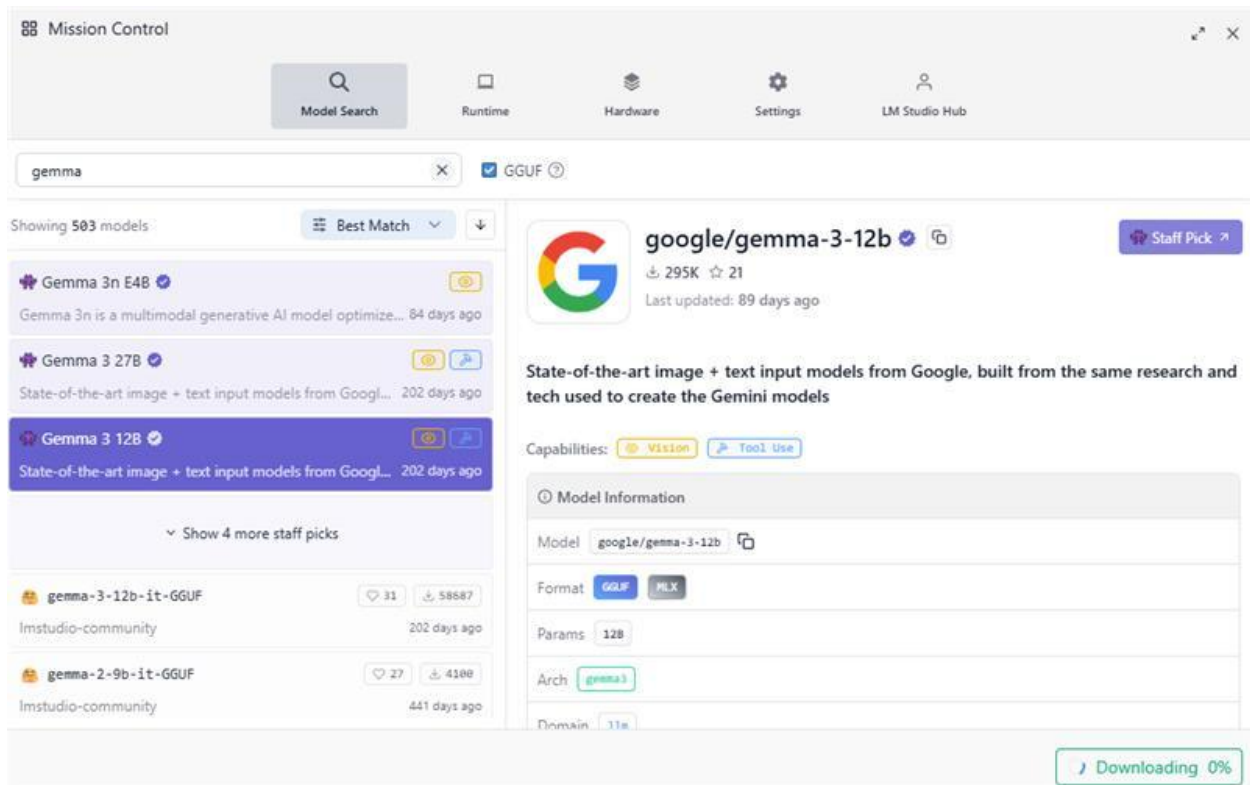


Developer

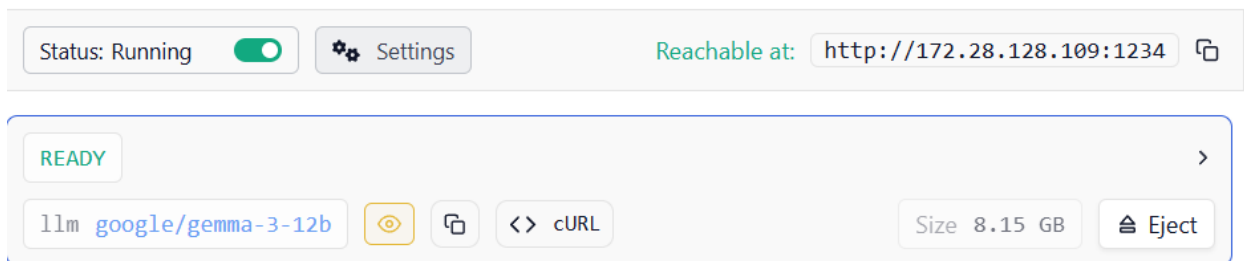
Show me everything

Continue →

Navigate to the search bar at the top and find an appropriate model to download based off your hardware



Navigate to the developer's tab to the left and make sure to run and load the model



Configure the model so that it serves the local network

Server Port ?	1234
Enable CORS ?	☐
Serve on Local Network ?	☒
Just-in-Time Model Loading ?	☒
Auto unload unused JIT loaded models ?	☒
Max idle TTL	60 minutes
Only Keep Last JIT Loaded Model ?	☒

To create a Web UI, code it or ask Claude AI using this prompt: “Code a Web-Based Interface connecting to LM studio at [http:// \(Computer’s IP Address\) :1234](http://(Computer’s IP Address):1234) using gradio and python and launch it automatically.”

YB Code a Web-Based Interface connecting to LM studio at <http://172.28.128.109:1234> using gradio and python and launch it automatically.

I'll create a Python script that uses Gradio to build a web interface connecting to your LM Studio server.

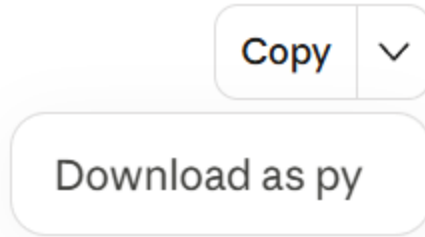
LM Studio Gradio Interface
Code



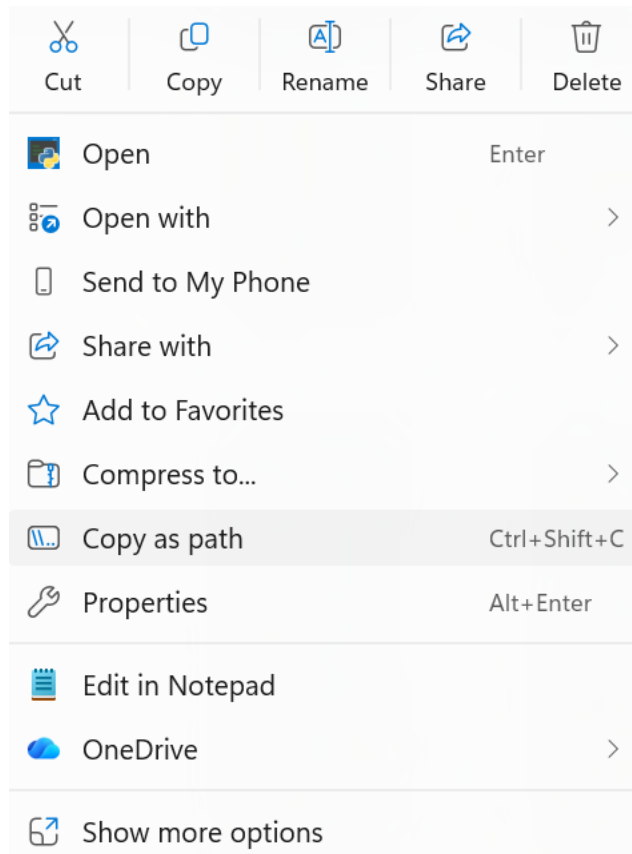
Install Python from Microsoft Store



Download the code as .py



Now go to your downloads folder, select the .py code file and copy the path



Now open PowerShell and enter: python (paste path)

```
PS C:\Users\Yuuki Burleigh> python "C:\Users\Yuuki Burleigh\Downloads\lm-studio-gradio.py"
```

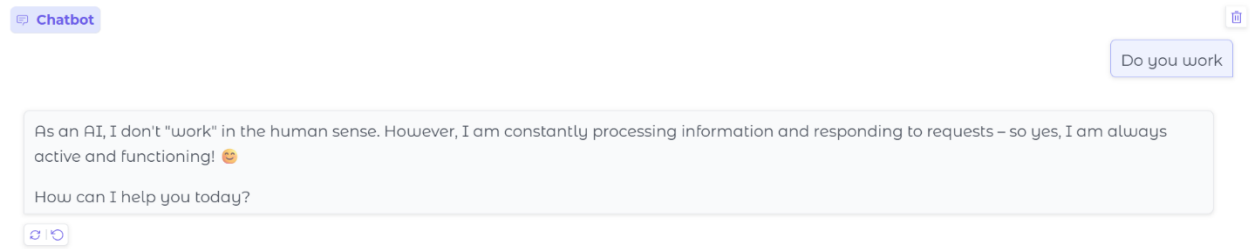
If PowerShell responds with an error stating “ModuleNotFoundError: No Module named ‘(name of module)’” as shown below.

```
PS C:\Users\kacin> python "C:\Users\kacin\Downloads\LM_Studio_UpdatedScript 1.py"
Traceback (most recent call last):
  File "C:\Users\kacin\Downloads\LM_Studio_UpdatedScript 1.py", line 6, in <module>
    import gradio as gr
ModuleNotFoundError: No module named 'gradio'
```

Enter the command pip install (name of the module) and rerun the python

```
PS C:\Users\Yuuki Burleigh> pip install gradio
```

Verify the AI is working by prompting



Problems

During this lab I encountered problems with trying to find an Open WebUI image to pull into Docker, but I was able to solve this by browsing through GitHub and learning the command to pull images into Docker.

Conclusion

Overall, this lab was very helpful in learning how to set up a Local and accessible AI for anyone within the same network to use as long as they know the IP address and port number of the Web Interface. This lab has helped build a strong foundation in AI so that I can build on top of the knowledge and experience I have gained by doing this lab for other future labs like training AI.